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An energy management method for combined load forecasting of fuel cell ships

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Abstract: The article proposes an energy management method (EMM) for fuel cell ships, which consists of two parts: load forecasting (LF) and real-time optimized scheduling (RTOS). LF is composed of Long Short-Term Memory neural networks, which are capable of being trained to predict future loads based on historical ship load information. RTOS consists of model predictive control and can allocate energy in real time. Initially, shiploads and battery status are collected in real time, and then LF passes the predicted load sequence to RTOS. RTOS optimizes real-time energy allocation based on the predicted load and the current operating status of two sets of batteries, while meeting system constraints and maintaining the battery SOC at a healthy level, to minimize operational cost consumption. The simulations demonstrate that the EMM can consistently maintain SOC at a healthy level and reduce fuel cell ship operating costs to some extent, thereby improving economic efficiency.

1. Introduction

Currently, the shipping industry plays a crucial role in facilitating global trade [1]. Existing ships mostly use internal combustion engines powered by fossil fuels, emitting large amounts of greenhouse gases such as carbon dioxide. It is estimated that by the middle of the 21st century, ships will emit 17% of the world's greenhouse gases [2]. In the context of the current “dual carbon” goals, there is an urgent need for ship technologies with zero greenhouse gas emissions.

In recent years, the application of fuel cells, a green energy source, to the transportation industry has received extensive research from scholars both domestically and internationally. However, due to their low rapid-response capabilities, fuel cells alone cannot meet the high dynamic power demands of ships. This necessitates the coordination with energy storage systems to guarantee stable navigation [3]. The key to efficient coordination between fuel cells and energy storage systems (ESS) lies in having an appropriate energy management method (EMM). Currently, the EMM of ships can mainly be classified into two categories: global planning and real-time optimization. Real-time optimization utilizes instantaneous online measurements of ship data to generate scheduling plans, which can better respond to the dynamic changes in shiploads compared to global planning, which relies on long-term historical data for offline analysis in advance. A three-tier real-time distributed power management system is proposed, which not only improves fuel efficiency and computational speed but also ensures sufficient energy reserves even during different sailing modes or faults [4]. Reinforcement learning methods are utilized for real-time energy scheduling of fuel cell ships, taking into account the dynamic degradation patterns of fuel cells and batteries [5]. To reduce tracking error and energy storage system losses, a model predictive control (MPC) method has been developed, and its effectiveness in power fluctuation mitigation and energy savings has been validated in HESS [6].



The above-mentioned EMMs all perform well under their respective objectives. However, the majority of real-time optimization methods lack a long-term perspective, which may result in the state of charge (SOC) of the energy storage equipment falling outside the healthy range. MPC has found extensive application in the transportation sector as well as other fields, owing to its capability to predict future steps [7]. However, current MPC models use a constant value to replace future load predictions, greatly weakening the performance of MPC.

Regarding this issue, a linear prediction method is employed, which not only results in imprecise prediction results but also has a prediction step size of only one step, making it impossible to make multi-step predictions and far from meeting the needs of MPC [8]. In [9], ARIMA was used for multi-step prediction of ship propulsion loads, but it was only applicable to regular changes in propulsion loads and could not handle the high volatility and uncertainty characteristic of actual ship load forecasting situations. Similarly, it is unable to meet the demands of MPC in practical applications.

In light of the aforementioned issues, this paper presents an EMM for fuel cell ships integrated with load prediction. The overall goal is to achieve the minimum operating cost while ensuring energy demand and sustaining an optimal SOC level of the battery. Specifically, future instantaneous loads predicted by Long Short-Term Memory neural networks (LSTM) and the current states of two sets of batteries are used to obtain the optimal energy allocation for the next time step by using MPC under certain constraints.

The rest parts of this paper are structured in the following. Section 2 introduces the fuel cell and lithium battery models, mainly FC and LB. Section 3 introduces the composition of the proposed energy management method. Section 4 conducts simulation verification and analysis of the proposed method based on the measured data of a fuel cell ship. Section 5 provides conclusions and future research directions.

2. Main equipment model

The main research objects of this article are fuel cells (FC), lithium batteries (LB), and shiploads in fuel cell ship power systems. The structure can be seen in Figure 1. The EMM predicts future loads based on historical load information and then collects operational status data from FC and LB for energy dispatch. FC and LB provide energy to the ship according to dispatch signals. Additionally, LB can also absorb energy from FC for self-charging purposes.

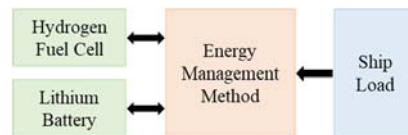


Figure 1. The structure of the fuel cell ship power system.

2.1. Fuel cell model

Figure 2 shows the efficiency of the FC. The rate at which hydrogen is consumed by the fuel cell $\dot{m}_{H_2}$ can be formulated as [10]:

$$\dot{m}_{H_2} = \frac{P_{FC}}{\eta_{FC} \cdot LHV_{H_2}} \quad (1)$$

where P_{FC} is the power of FC, η_{FC} is the efficiency of FC, and LHV_{H_2} is the low calorific value of hydrogen.

The operating cost of fuel cells C_{FC} can be expressed as:

$$C_{FC} = C_{H_2} \cdot \Delta m_{H_2} \quad (2)$$

where C_{H_2} is the price of hydrogen and Δm_{H_2} is the mass of hydrogen consumed by FC.

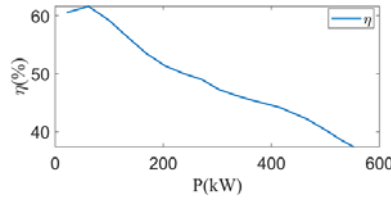


Figure 2. The efficiency of FC.

2.2. Lithium battery model

As the energy storage system of the fuel cell ships, the output power P_{LB} of the LB can be formulated as:

$$P_{LB} = (U_{oc} - R_{LB} I_{LB}) I_{LB} \quad (3)$$

where U_{oc} is the voltage of LB, R_{LB} is the internal resistance of LB, and I_{LB} is the current of LB.

The SOC of the LB is the proportion of the actual remaining charge to nominal capacity, which can be calculated by using Equation (4):

$$\dot{SOC} = -\frac{U_{oc} - \sqrt{U_{oc}^2 - 4P_{LB}R_{LB}}}{2R_{LB}Q_e} \quad (4)$$

where Q_e is the nominal charge value of LB.

In accordance with the equivalent consumption method, the hydrogen consumption of LB can be calculated through Equation (5):

$$\dot{m}_{LB}^{eq} = \gamma \cdot \frac{\alpha \cdot P_{LB}(h)}{LHV_{H_2}} \quad (5)$$

$$\alpha = \begin{cases} \frac{1}{\eta_{dis}\eta_{chr}\eta_{DC/DC}}, & P_{LB} \geq 0 \\ \frac{\eta_{chr}}{\eta_{dis}\eta_{DC/DC}}, & P_{LB} < 0 \end{cases} \quad (6)$$

where $\dot{m}_{LB}^{eq}$ is the rate of equivalent hydrogen consumption by LB, γ is the balance parameter, SOC_{max} and SOC_{min} are the upper and lower bounds of the SOC constraints, and $\eta_{DC/DC}$, η_{chr} , and η_{dis} are the efficiency of DC/DC converters, charging and discharge efficiency of LB.

So, the operating cost C_{LB} of the LB can be expressed as:

$$C_{LB} = C_{H_2} \cdot \Delta m_{LB}^{eq} \quad (7)$$

where Δm_{LB}^{eq} is the equivalent hydrogen mass consumed by LB.

3. Energy management method

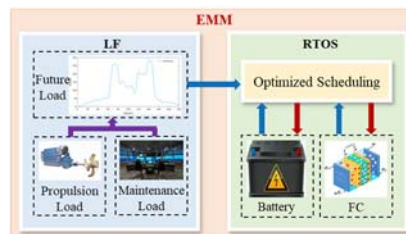


Figure 3. The structure of EMM.

The frame of the EMM outlined in this paper is depicted in Figure 3, primarily consisting of two parts: load forecasting (LF) and real-time optimized scheduling (RTOS). LF collects historical load information, including propulsion load and maintenance load, to predict shiploads for a future period.

OS uses the future load provided by ILF, along with the current status of FC (output power and hydrogen consumption) and LB (state of charge), to determine the optimal energy allocation between FC and LB, achieving the minimum total operating cost and a healthy SOC level.

3.1. Instantaneous load forecasting

Compared to traditional machine learning models that can only utilize the current time step or a few recent time steps, LSTM is effective in capturing and maintaining long-term dependencies in sequential data. It can automatically learn and adjust the weights of memory units based on the characteristics of input data, allowing them to better adapt to the uncertainty and volatility of shiploads. The particular structure is depicted in Figure 4, where the yellow box is the input gate, controlling data input; the purple box is the output gate, jointly determining the current output with the memory unit from the previous time step; the blue box represents the forget gate, deciding how much information from previous time step is forgotten.

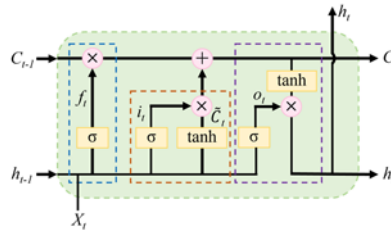


Figure 4. The structure of LSTM.

3.2. Model predictive control

MPC, as a widely applied real-time optimization algorithm, can predict future system behavior and integrate seamlessly with shipload forecasting for energy scheduling. MPC effectively adapts to varying demands in shiploads, utilizing real-time equipment status and predicted load information to solve the objective function within restrictions. This capability is crucial for real-time energy scheduling on ships. The hydrogen consumption rate $\dot{m}_{H_2}$ of FC and the SOC of LB are state variables of MPC, and P_{HC} and P_{bat} are control and output variables. The optimization objective of real-time scheduling is to reduce the overall operating cost, so the cost function and restrictions of MPC can be expressed as:

$$J = C_{total} = \sum_{h=1}^{N_p} (C_{FC}(h) + C_{LB}(h)) \quad (8)$$

$$\begin{aligned} s.t. \quad & P_{FC}(h) + P_{LB}(h) = P_{load}(h) \\ & P_{FC}(h) \in [P_{FC.min}, P_{FC.max}] \\ & P_{LB}(h) \in [P_{LB.min}, P_{LB.max}] \\ & SOC(h) \in [SOC_{min}, SOC_{max}] \\ & |\Delta P_{FC}(h)| \leq \Delta P_{FC.max} \end{aligned} \quad (9)$$

where N_p is the prediction step size, P_{load} is the predicted load output by LF at the corresponding time, $P_{FC.min}$ and $P_{FC.max}$ are the upper and lower bounds of FC output power, $P_{LB.min}$ and $P_{LB.max}$ are the upper and lower bounds of LB output power, and ΔP_{FC} and $\Delta P_{FC.max}$ are the rate of FC output power change and the maximum allowable rate of change.

4. Simulations and results analysis

In this section, based on operational data from a domestically developed hydrogen fuel cell ship in China, “Three Gorges Hydrogen Boat 1”, which ran for 10 hours, the proposed EMM is validated via simulation. This ship is China’s first CCS-class fuel cell ship. The dataset includes measured output power of FC and LB, as well as shipload, sampled at a frequency of 1 second. The primary parameters

of the “Three Gorges Hydrogen Boat 1” are depicted in Table 1. Due to confidentiality agreements, further detailed information cannot be provided.

Table 1. The primary parameters of the “Three Gorges Hydrogen Boat 1”.

Parameter	Value	Unit
Length/Width/Height	49.9/10.4/3.2	m
Maximum speed	28	km/h
[SOC _{min} , SOC _{max}]	[50%, 80%]	-
$\Delta P_{HFC,max}$	48	kW
C_{H_2}	4	USD/kg

4.1. Instantaneous load forecasting

Based on actual forecasting performance and the speed of the model, in this paper, LSTM gathers historical load information from the past 30 seconds as input data. After completing training, it outputs values for the next N_P time steps in one go. In this study, N_P is set to 5, and LSTM performs forward-rolling prediction. The 200-second forecast results are illustrated in Figure 5. The load profile in this paragraph includes three main operational conditions of ships: acceleration, deceleration, and operations. The average absolute error is calculated to be 6.49 kW and the root mean square error is 10.11 kW, which performs well.

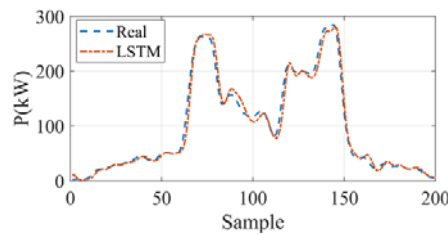


Figure 5. Prediction results.

4.2. Real-time optimization scheduling

Using the power shown in Figure 5 as the shipload for simulation, the power distribution results obtained are illustrated in Figure 6. The sum of the output power of the LB and FC is equal to the shipload. During the operation of the ship, the FC provides the main energy. When the load fluctuates at high frequencies, the battery provides high-frequency power and can absorb the remaining energy output by the fuel cell, maintaining the health of its own SOC.

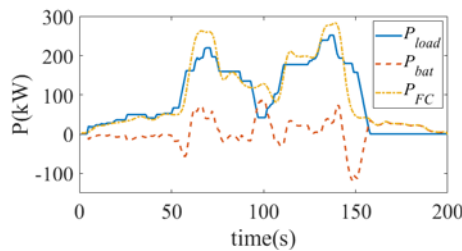


Figure 6. Energy allocation results.

The operating costs of the proposed EMM and the actual energy management strategy are shown in Table 2. Within a running time of 200 seconds, the total operating cost decreased by about 4%, which is 0.0159 USD. Meanwhile, Figure 7 depicts the variation curve of battery SOC during actual navigation over 10 hours. This is mainly because the actual energy management method is not reasonable, with FC and LB bearing almost equal responsibilities. The proposed EMM aligns with practical needs, where FC serves as the primary energy source and LB is the auxiliary energy source. This approach, with a relatively higher CFC, significantly reduces CLB, thereby lowering total costs.

It can be observed that the actual SOC fluctuates significantly and is unstable. In contrast, the SOC curve of the proposed method starts declining after the ship starts, fluctuating above 50% throughout, ensuring a healthy SOC level that is conducive to prolonged battery life.

Table 2. Operating costs.

Method	C_{FC}	C_{bat}	C_{total}
Real method	0.2138	0.1856	0.3994
Proposed method	0.3244	0.0592	0.3835

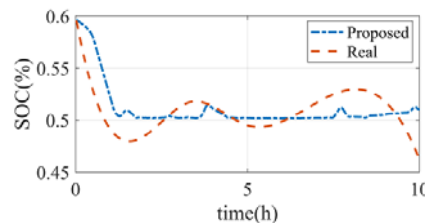


Figure 7. The variation curve of SOC.

5. Conclusion

This paper proposes an EMM for fuel cell ships' combined load forecasting. The proposed method consists of two parts: LF and RTOS. LF predicts the load for future time steps, while RTOS performs real-time energy allocation based on the predicted load from LF and the current operating status of FC and LB while ensuring compliance with system restrictions. The simulation validate that the proposed method can reduce operational costs and improve economic efficiency compared to actual energy management strategies to some extent. Moreover, it ensures that the SOC of LB remains in a healthy state. This study offers valuable perspectives for the development of fuel cell ships.

In the future, efforts will focus on finding more precise and efficient load forecasting methods to enhance the accuracy and speed of predictions. Additionally, attention will be given to the lifespan of energy supply equipment and dynamic changes during actual operation, aiming to refine the EMM.

Acknowledgments

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References

- [1] Z. Zhang, C. Guan, H. Chen, X. Yang, W. Gong and A. Yang. "Adaptive Privacy-Preserving Federated Learning for Fault Diagnosis in Internet of Ships". In IEEE Internet of Things Journal, Vol. 9, No. 9, pp. 6, 844-6, 854, May 1, 2022.
- [2] Y. Sun, Q. Xu, Y. Yuan and B. Yang. Optimal Energy Management of Fuel Cell Hybrid Electric Ships Considering Fuel Cell Aging Cost. In 2020 IEEE/IAS Industrial and Commercial Power System Asia (I&CPS Asia), Jul. 2020, pp. 240-245.
- [3] DNV, "Fuel cells for ships", Research and Innovation, DNV, Oslo, Norway, Position Tech. Rep. Paper 13, 2012.
- [4] P. Xie et al. "A distributed real-time power management scheme for shipboard zonal multi-microgrid system", Appl. Energy, Vol. 317, 2022, Art. No. 119, 072.
- [5] N. P. Reddy, R. Skjetne, O. S. Os and D. Papageorgiou. A Comparison of the State-of-the-Art Reinforcement Learning Algorithms for Health-Aware Energy and Emissions Management in Zero-Emission Ships, IEEE J. Emerg. Sel. Top. Ind. Electron., Vol. 5, No. 1, pp. 149-166, Jan. 2024.
- [6] J. Hou, J. Sun and H. Hofmann. "Control development and performance evaluation for battery/flywheel hybrid energy storage solutions to mitigate load fluctuations in all-electric ship propulsion systems", Appl. Energy, Vol. 212, pp. 919-930, 2018.

- [7] W. Chen et al. Robust Real-Time Shipboard Energy Management System with Improved Adaptive Model Predictive Control, IEEE Access, Vol. 11, pp. 110, 342-110, 360, 2023.
- [8] I. Kolmanovsky, I. Sivergina and J. Sun. "Simultaneous input and parameter estimation with input observers and set-membership parameter bounding: Theory and an automotive application", Int. J. Adaptive Control Signal Process, Vol. 20, No. 5, pp. 225-246, 2006.
- [9] P. Xie, S. Tan, N. Bazmohammadi, J. M. Guerrero and J. C. Vasquez. A Real-Time Power Management Strategy for Hybrid Electrical Ships Under Highly Fluctuated Propulsion Loads, IEEE Syst. J., Vol. 17, No. 1, pp. 395-406, Mar. 2023.
- [10] S. Quan, Y. X. Wang, X. Xiao, H. He and F. Sun. Real-time energy management for fuel cell electric vehicle using speed prediction-based model predictive control considering performance degradation, Appl. Energy, Vol. 304, p. 117, 845, Dec. 2021.